

SBC 201 Chapter 10 Means of Egress — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 1–20 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 10, source file Reference\SBC 201 2024\source_reference\Chapter_10 – MEANS OF EGRESS.txt.
- **Extraction audit:** Skill-finetune re-run. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **645** independently checkable numeric records (**557** Verified, **88** Verify source). Unresolved OCR is listed in the register and is not a design-release value. Pre-skill baseline retained as ..._Cursor-Grok-4.6-pre-skill.md.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, fire-engineering report, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from an outbound code, standard, commentary example, or the existing chapter summary. Where Chapter 10 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications.
2. The exact Riyadh AHJ/permit pathway, project stage, fire-strategy status and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Automatic sprinkler protection is not selected. **NFPA 13 / Section 903.3.1.1** and **NFPA 13R / Section 903.3.1.2** are shown as separate branches wherever Chapter 10 differentiates or jointly recognizes them.
4. Emergency voice/alarm communication system (EVACS) compliance with Section 907.5.2.2 is unconfirmed.
5. Building height, storey count, grade plane, accessible floors and occupied-roof configuration, level(s) of exit discharge, unit type, occupant loads, mixed uses and amenity layouts are unconfirmed.
6. Table 1004.5 Residential, Table 1006.2.1, Tables 1006.3.4(1)/(2), Table 1017.2 and both OCR-labelled Table 1020.2 extracts contain source segmentation or footnote uncertainty. Their affected rows are marked **Verify source** and are not design release values.
7. Accessibility technical geometry cross-referenced to Chapter 11/ICC A117.1, fire-protection systems in Chapter 9, high-rise provisions in Section 403, smokeproof-enclosure systems in Section 909.20, and other outbound standards require separate verification.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, use, occupant load, sprinkler branch or exception exists.
Not typical	Not expected in a conventional R-2 tower, but retain if the feature appears in podium, amenity, roof or mixed-use areas.
External verification	Chapter 10 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 10 source text or an unambiguous table cell.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 10 application	Required project action
Sprinkler basis	Unconfirmed: neither NFPA 13 nor NFPA 13R is assumed	Controls capacity reductions, separation, common path, travel-distance table branch, dead ends, accessible-egress exceptions and corridor table branch	Fire engineer and code consultant to lock the Chapter 9 system and record whether Section 903.3.1.1 or 903.3.1.2 applies; verify high-rise permissibility outside Chapter 10
EVACS	Unconfirmed	Reduced stair capacity factor requires both qualifying sprinklers and Section 907.5.2.2 EVACS	Fire alarm designer to confirm EVACS scope and cause/effect; until then retain base capacity factors
Storeys and height	Occupied floor stated above 23 m; exact height, grade plane and storey count unconfirmed	Controls high-rise outbound provisions, accessible-egress elevator trigger, enclosure rating, roof access and stair signs	Issue a signed code datum sheet with grade plane, building height, occupied-floor elevations, basements, mezzanines and storey count
Per-storey occupant load	Unconfirmed	Controls number of exits/accesses to exits and cumulative stair/discharge sizing	Produce a storey-by-storey occupant-load and egress-routing schedule
Per-space occupant load	Unconfirmed	Controls one-door logic, common path, door swing, panic hardware and assembly rules	Produce a room/space load schedule using actual function, net/gross basis and layout
Unit type	Unconfirmed: dwelling units versus sleeping units	Controls Tables 1006.3.4(1)/(2), internal door exceptions and single-exit logic	Classify every R-2 unit type and show the applicable table branch on life-safety plans
Podium and mixed uses	Detailed uses unconfirmed	Shared egress must satisfy the most stringent served occupancy; residential rules cannot be applied to retail, parking, business or assembly by default	Freeze occupancy/use by room and fire area; identify every shared egress component
Amenity assembly	Configuration, fixed seating and occupant load unconfirmed	Triggers Table 1004.5 assembly factors, Section 1030 aisle/seating rules, possible main-exit provisions and occupant-load posting	Classify each lounge, multipurpose room, gym, pool deck, cinema and roof amenity by actual function and seating arrangement
Accessible floors and occupied roof	Unconfirmed	Controls number and components of accessible means of egress and the four-storey elevator trigger	Accessibility consultant to map all required accessible floors/roof and the complete route to a public way
Level of exit discharge	Unconfirmed, including possible multiple discharge levels	Controls elevator trigger datum, stair termination, discharge exceptions, accessible discharge and convergence	Mark each level of exit discharge in section and trace every exit independently to the public way

Decision / gap	Current project basis	Why it controls Chapter 10 application	Required project action
NOC and fire strategy	SCD NOC and stamped fire-strategy status unconfirmed	Authority acceptance, smoke control, active systems, assisted evacuation and high-rise strategy cannot be concluded from Chapter 10 alone	Engage the qualified local/fire consultant; align life-safety drawings, fire strategy, SBPS submission and SCD comments before design freeze

4. General egress baseline

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1003.2	General egress ceiling height	Means of egress ceiling height not less than 2300 mm above finished floor	All egress components	Dwelling/sleeping-unit ceilings, stair headroom, door height, ramp headroom and other listed cases are separately controlled	Direct	Coordinate reflected ceilings, services and signage to preserve the applicable clear height	Verified
1003.3.1	Protruding-object headroom	Minimum headroom 2000 mm over circulation paths; not more than 50% of the egress ceiling area may be reduced	Protrusions below the general ceiling height	Door closers and stops may reduce headroom only to 2000 mm	Direct	Model clear headroom along all walking lines, not only corridor centerlines	Verified
1003.3.1	Low-headroom barrier	Barrier required where clearance is less than 2000 mm ; leading edge 700 mm maximum above floor	Any low-clearance circulation path	No numeric branch stated	Direct	Detail cane-detectable protection below stairs and services	Verified
1003.3.2–1003.3.3	Protruding objects	Post-mounted overhang maximum 100 mm where leading edge is over 700 mm and under 2000 mm ; where post spacing exceeds 300 mm , obstruction bottom is 700 mm maximum or 2000 mm minimum ; horizontal projection maximum 100 mm between 700–2000 mm high	Object projects into a circulation path	Handrail exception permits 115 mm projection; sloping handrail portions between stair/ramp endpoints are excepted as stated	Direct	Audit wall- and post-mounted signs, cabinets and equipment against cane/head zones	Verified
1003.5	Elevation changes	Changes under 300 mm use a sloped surface; slopes over 1:20 (5%) use a 1012 ramp; change 150 mm or less requires handrails or contrasting floor finish	Elevation change in means of egress	Listed nonaccessible exceptions include a single riser or two risers with 325 mm minimum tread and at least one handrail within 750 mm of travel centerline	Direct	Eliminate isolated steps from common egress; document any exact exception	Verified
1003.6	Egress continuity	Path shall not be interrupted; minimum width or required capacity shall not diminish along travel	Entire exit access, exit and exit discharge	Only Chapter 10 permitted projections/components	Direct	Trace the controlling width continuously to the public way	Verified

5. Occupant load and capacity

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1004.2.1	Cumulative load through intervening spaces	Capacity is based on the combined occupant load of interconnected accessory/intervening spaces to each point along the path	Occupants egress through another room/area/space	Do not double-count occupants with independent routes	Direct	Build a cumulative load tree for lobbies, corridors and discharge paths	Verified
1004.2.3	Loads from adjacent stories	Occupant loads from separate stories are not added , except for convergence under 1005.6	Vertical egress sizing by story	Mezzanine loads and convergence follow their own rules	Direct	Size story egress separately, then test convergence	Verified
1004.3	Multiple functions in one area	Calculate each function independently using its applicable occupant-load factor	Mixed-function amenity or podium space	Net areas are removed before applying a gross-area factor to the balance	Conditional	Zone each actual function on the occupant-load plan	Verified
1004.4	Multiple occupancies	Apply means-of-egress rules to each occupancy portion; shared components meet the most stringent requirement of occupancies served	Podium or mixed-use tower	No residential default for shared paths	Conditional	Tag shared corridors, stairs and discharge with all served occupancies	Verified
1004.7	Outdoor areas and occupied roofs	Building official assigns occupant load from anticipated use. Where additional persons use a yard, patio, occupied roof, court or similar area and its egress passes through the building, building egress is based on the sum of building and outdoor-area occupant loads	Outdoor area accessible to and usable by building occupants	The aggregation rule does not apply to outdoor areas associated with an individual Group R-2 dwelling unit; service-only outdoor areas need only one means	Conditional	Include shared roof/terrace amenity loads in the cumulative route; keep private unit balconies/patios on the stated exception branch	Verified
Table 1004.5	Areas without fixed seating	Occupant load is not less than floor area divided by the factor for the function of space	Every non-fixed-seat space	Building official may approve actual load lower than calculated; approval is required	Direct	Record factor, net/gross basis, area and rounded load per space	Verified
Table 1004.5	Residential factor	Residential row exists, but numeric factor and gross/net basis cannot be unambiguously segmented from the OCR table	R-2 dwelling/sleeping areas	No numeric value is adopted in this tier	External verification	Verify published Table 1004.5 before calculating residential loads	Verify source
Table 1004.5	Likely tower support functions	Business 14 m² gross/person ; commercial kitchen 19 m² gross/person ; exercise room 4.6 m² gross/person ; parking garage 19 m² gross/person ; accessory storage/mechanical 28 m² gross/person	Where those actual functions occur	Use actual function, not the overall R-2 group label	Conditional	Apply only to mapped rooms with matching functions	Verified
Table 1004.5	Assembly amenity factors	Standing 0.46 m² net/person ; concentrated chairs only 0.65 m² net/person ; unconcentrated tables/chairs 1.4 m² net/person	Amenity used for assembly without fixed seating	Pools/decks rows remain OCR-unresolved and are not supplied here	Conditional	Test every intended furniture/event layout and use the governing load	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1004.5.1	Increased occupant load	Modified load permitted if all other code requirements use it and density does not exceed 1 person per 0.65 m² of occupiable floor space	Designer/owner proposes higher load	All affected provisions must use modified load	Conditional	Record the approved design load consistently across all disciplines	Verified
1004.6	Fixed seating	Without dividing arms: 1 person per 450 mm of seating length; booths: 1 person per 600 mm measured at backrest	Fixed seating in amenity/assembly areas	Wheelchair and companion spaces each count as one occupant	Conditional	Count seats and continuous benches from the actual seating plan	Verified
1004.8	Concentrated business	Approved actual load, but not less than 1 person per 4.65 m² gross	Call centres, trading floors, data-processing centres and similar dense business uses	Building-official approval required	Conditional	Apply only if such a podium/management use is confirmed	Verified
1005.3.1	Base stair capacity	Stair capacity = served occupant load × 7.6 mm/person	All stairs unless an exception is proven	Assembly seating has separate listed branches	Direct	Retain base factor until sprinkler and EVACS evidence is locked	Verified
1005.3.1 Ex. 1	Reduced stair capacity	5.08 mm/person	Other than H/I-2, and building throughout has Section 903.3.1.1 or 903.3.1.2 sprinklers and Section 907.5.2.2 EVACS	NFPA 13 and NFPA 13R both satisfy the Chapter 10 sprinkler wording; EVACS is additionally mandatory	Conditional	Do not use reduced factor unless both system conditions are documented	Verified
1005.3.2	Base non-stair capacity	Other egress component capacity = served load × 5.08 mm/person	Doors, corridors, ramps and other non-stair components	Reduced-factor Exception 1 is OCR-corrupted	Direct	Use 5.08 mm/person ; do not infer a reduced value	Verified
1005.3.2 Ex. 1	Reduced non-stair capacity	Mandatory OCR is corrupted; numeric factor is not recoverable from the supplied extract	Proposed reduced non-stair factor	No value adopted from commentary or outbound source	External verification	Verify the published clause before any reduction	Verify source
1005.6	Egress convergence	After convergence from stories above/below, provide the larger of the largest minimum width or the sum of required capacities for the two adjacent stories	Stairs/ramps converge at an intermediate level	Limited to the two adjacent stories described	Conditional	Check podium/ground convergence before fixing lobby width	Verified
1005.7 / 1005.7.1–1005.7.3	Encroachments and projections	Fully opened door reduces required width by maximum 175 mm ; door in any position by maximum one-half ; other trim/decorative projection maximum 40 mm each side ; protruding objects comply with 1003.3	Door, hardware or projection enters required egress width	Surface latch hardware exception requires mounting 850–1200 mm high and stated orientation; door-swing restriction does not apply inside individual R-2 dwelling/sleeping units	Direct	Overlay all door swings and wall projections on net-width plans	Verified

6. Exit counts and loss-of-exit capacity

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1006.2.1	Space-level two-way-out trigger	Provide two exits or exit access doorways from a space where occupant load or common path exceeds Table 1006.2.1	Applied to each room/space	Mechanical penthouse common-path measurement and other listed exceptions	Direct	Check each unit, amenity, plant and tenant space independently	Verified
1006.2.1.1	Large-space exit count	Space load 501–1,000: 3 exits/access doorways; greater than 1,000: 4	Individual space exceeds 500 occupants	Separate from story-level Table 1006.3.3	Conditional	Apply to large podium/amenity spaces if present	Verified
1006.3 / 1006.3.1	Stories and occupied roofs	Number of separate and distinct exits/accesses is based on aggregate load served; where stairs serve multiple stories or a story and occupied roof, use each story or occupied-roof load individually to calculate its required exits/accesses	Every story and occupied roof	Mezzanine and convergence rules remain separate	Direct	Add each occupied roof as its own line in the exit-count schedule	Verified
Table 1006.3.3	Story/occupied-roof exit count	Load 1–500: 2; 501–1,000: 3; more than 1,000: 4 exits or accesses to exits	Each story and occupied roof	Single-exit permissions under 1006.3.4 are separate exceptions; required count is maintained to exit discharge/public way	Direct	Schedule roof and story counts independently from every space/unit check	Verified
1006.3.4 Item 5	Unit-level one-door logic	An individual single- or multistorey dwelling unit may have a single exit/access if it complies with 1006.2.1 and, outside the unit entrance, access is to at least two approved independent exits , unless the unit exit discharges directly outside at the level of discharge	Individual dwelling unit	This permission does not reduce the required number of exits from the story	Conditional	Show one unit entrance door and two independent story exits as separate checks	Verified
Table 1006.2.1	R-2 one-door space limit	OCR/inventory reads maximum 20 occupants and sprinklered common path 38 m ; unsprinklered branch reads Not Permitted	R-2 room/space with one exit access doorway	Table note recognizes Section 903.3.1.1 or 903.3.1.2; table segmentation requires published-source check	Conditional	Treat as a hold point; verify table before relying on a one-door unit/space	Verify source
Table 1006.3.4(1)	Single-exit R-2 dwelling story	OCR/inventory reads basement through third story above grade: maximum 4 dwelling units and 38 m travel	R-2 consisting of dwelling units	Requires Section 903.3.1.1 or 903.3.1.2 sprinklers and Section 1031 EEROs; fourth story and higher reads Not Permitted	Conditional	High-rise residential floors should retain multiple story exits; verify low podium/residential levels separately	Verify source
Table 1006.3.4(2)	R-2 sleeping-unit story branch	Table is for R-2 sleeping units; inventory rows are flattened and upper stories read single exit Not Permitted from the third story and higher	R-2 consisting of sleeping units	Exact row/occupancy attachment requires published-source check	Conditional	Confirm unit type; do not transpose dwelling-unit limits to sleeping units	Verify source
1005.5	Exit-loss check	Remaining capacity/width after loss of one exit must be at least 50% required	Any floor/space requiring multiple exits	Applies alongside exit-count minimums	Direct	Distribute capacity so one oversized/critical exit does not fail the system	Verified

7. Common path, exit access travel and exit separation

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
Table 1006.2.1	R-2 common path	OCR/inventory reads 38 m with qualifying sprinklers and Not Permitted without	One-door R-2 space branch	Note a recognizes Section 903.3.1.1 or 903.3.1.2; table OCR requires verification	Conditional	Verify published table and measure from the most remote point to choice of two routes	Verify source
Table 1017.2	Group R exit access travel	Flattened OCR row for A/E/F-1/I-1/M/R/S-1 reads “ 60° / 75° ”; limits/footnote markers are unresolved	Total exit access travel for Group R	Sprinkler footnote references Section 903.3.1.1 or 903.3.1.2	External verification	Do not use 60/75 as released design criteria until published table is checked	Verify source
1017.3	Measurement to exit	Where more than one exit is required, measure travel distance to the nearest exit	Multiple exits required	Listed special measurement exceptions	Direct	Show measured travel paths, not straight-line radii	Verified
1007.1.1	Two-exit remoteness — base	Separation at least one-half the maximum overall diagonal of the building/area served	Two exits/access doorways/stairs/ramps required	Straight-line measurement unless corridor method applies	Direct	Dimension the governing diagonal and exit separation on each plan	Verified
1007.1.1	Two-exit remoteness — sprinkler branch	Separation may be at least one-third the maximum overall diagonal	Building throughout sprinklered under 903.3.1.1 or 903.3.1.2	No selection between NFPA 13 and 13R within this Chapter 10 clause	Conditional	Use one-third only after sprinkler basis is formally fixed	Verified
1007.1.1	Interconnected/scissor stairs	Interlocking or scissor stairways count as one exit stairway	Scissor/interlocking core proposed	Does not supply two required exits by itself	Conditional	Do not count two flights in one scissor enclosure as two exits	Verified
1007.1.1	Corridor separation measurement	For stairs/ramps interconnected by a 1-hour corridor complying with 1020, measure along the shortest direct line within the corridor	Rated interconnecting corridor arrangement	Corridor must satisfy the stated protection	Conditional	Identify exact measurement method in fire strategy	Verified
1007.1.2	Three-or-more exits	At least two of three or more required exits/access doorways must satisfy 1007.1.1 remoteness	Story/space requires 3 or 4 exits	Remaining exits still must be arranged to remain available	Conditional	Flag the remote pair in the egress schedule	Verified
1020.5	Dead end — base	Where more than one exit/access doorway is required, dead-end corridor maximum 6 m	Multiple-exit condition	Geometric exception where dead end is less than 2.5 × least corridor width	Direct	Use 6 m unless a listed exception is demonstrated	Verified
1020.5 Ex. 2	Dead end — NFPA 13 branch	Group R-2 dead-end corridor maximum 15 m	Building throughout sprinklered under 903.3.1.1 only	NFPA 13R / 903.3.1.2 is not listed in this exception	Conditional	Keep separate 13 and 13R plan checks; do not give 13R the 15 m allowance	Verified
1017.2.1	Exterior egress balcony increase	Table 1017.2 travel may increase by up to 30 m where the last portion occurs on a 1021-compliant exterior egress balcony	Exterior egress balcony used	Base Table 1017.2 value still requires verification	Conditional	Treat only as a documented alternative after balcony compliance and base limit are verified	Verified

8. Accessible means of egress

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1009.1	Minimum accessible means	Accessible spaces require at least one accessible means of egress	Any required accessible space	Accessible mezzanine and assembly exceptions listed	Direct	Map a continuous accessible egress path to the public way	Verified
1009.1	Multiple accessible means	Where more than one means of egress is required from an accessible space, each accessible portion requires at least two accessible means of egress	1006.2/1006.3 requires multiple means	Assembly ramped/stepped-aisle exception may allow one when common path is accessible and satisfies 1030.8	Direct	Identify the two independent accessible options on every applicable floor	Verified
1009.2	Permitted components	Components include accessible routes, compliant interior/exterior/exit-access stairs, elevators, platform lifts, horizontal exits, ramps, areas of refuge and exterior assisted-rescue areas	Building the complete accessible route	Each component must satisfy its cited section and outbound accessibility scoping	Direct	Create a component-by-component accessible egress diagram	Verified
1009.2.1	Elevator trigger	Where a required accessible floor or occupied roof is four or more stories above or below a level of exit discharge, at least one required accessible means shall be a 1009.4 elevator	Four-storey vertical separation	OCR duplicated wording in one exception; base requirement is clear but inventory remains flagged	Conditional	Reserve an accessible-egress elevator strategy unless an exception is proven	Verify source
1009.2.1 Ex. 1	Elevator exception — horizontal exit	Elevator not required on floors at/above discharge that have a horizontal exit	Building throughout sprinklered under 903.3.1.1 or 903.3.1.2	Does not apply to floors below discharge	Conditional	Demonstrate horizontal-exit refuge, capacity and discharge before using exception	Verify source
1009.2.1 Ex. 2	Elevator exception — ramp	Elevator not required on floors provided with a 1012-compliant ramp	Building throughout sprinklered under 903.3.1.1 or 903.3.1.2	Ramp must serve the applicable levels	Conditional	Apply only to an actual continuous ramp solution	Verify source
1009.3.2	Accessible-egress stair width	Clear width 1200 mm minimum between handrails	Stair used as accessible means of egress	Width not required with 903.3.1.1 or 903.3.1.2 sprinklers, or from refuge area with horizontal exit	Conditional	Check clear width between handrails separately from nominal stair width	Verified
1009.3.3 Ex. 5	R-2 stair area of refuge	Areas of refuge are not required at stairways in Group R-2 occupancies	Stair forms part of an accessible means of egress in R-2	This is only an area-of-refuge exception; it does not remove the accessible-means-of-egress requirement or the accessible route to the stair/elevator	Conditional	Continue to provide the required accessible means, stair access and other applicable communication/elevator components	Verified
1009.6.3	Area-of-refuge wheelchair spaces	One 750 × 1300 mm wheelchair space per 200 occupants or portion served	Area of refuge provided/required	Other protection and communication provisions also apply	Conditional	Lay out spaces without reducing egress width	Verified
1009.7.2–1009.7.4	Exterior assisted-rescue area	Separation 1 hour ; rated wall extends 3000 mm vertically and 3 m horizontally each side, or 1200 mm perpendicular alternative; other sides at least 50% open ; stair 1200 mm clear between handrails	Exterior assisted-rescue option at level of discharge	Rating/width exceptions where throughout sprinklered under 903.3.1.1 or 903.3.1.2	Conditional	Coordinate façade exposure, opening protection and landing geometry	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1009.8	Two-way communication	Provide at each elevator/bank landing on each accessible floor one or more stories above or below discharge	Accessible elevator landings	Areas of refuge, ramps, service/freight/private residence elevators and listed institutional cases have exceptions	Conditional	Include devices, signage, monitoring location and power/interface in fire strategy	Verified

9. Illumination and signs

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1008.2	Illumination while occupied	Means of egress serving a room/space is illuminated whenever occupied	Occupied room or space	Dwelling units and sleeping units in Group R-2 are excepted from this continuous-illumination rule; the exception does not extend to required common egress outside the unit	Direct	Separate in-unit lighting controls from continuously available common egress lighting	Verified
1008.2.1	Normal common-egress illumination	At least 11 lux at walking surface	Required common means of egress outside the excepted R-2 dwelling/sleeping unit	Assembly performance exception is separate	Direct	Put 11-lux minimum on common corridors, exits and discharge criteria	Verified
1008.2.1	Common stair illumination	At least 110 lux at walking surface on exit-access stairs, exit stairs and required landings when in use	Required common stairs outside the excepted R-2 dwelling/sleeping unit	Assembly performance reduction may reach 2.2 lux only with automatic restoration and listed conditions	Direct	Coordinate normal controls so common stairs meet 110 lux when used	Verified
1008.3.2	Emergency-illuminated components	Emergency system automatically illuminates interior exit-access stairs/ramps, interior/exterior exit stairs/ramps, exit passageways, discharge vestibules/areas and exterior exit landings	Building requires two or more exits/accesses	Listed component scope is cumulative	Direct	Include the complete route, not only stairs and exit signs	Verified
1008.3.3 Item 5	Large public-restroom emergency lighting	Public restrooms over 28 m² are automatically illuminated by the emergency electrical system upon power failure	Public restroom area exceeds 28 m²	Other listed emergency-response rooms have no dimensional trigger in this clause	Conditional	Add qualifying public restrooms to emergency-lighting schedules and calculations	Verified
1008.3.4	Emergency duration	Emergency power for at least 90 minutes	Emergency egress lighting	Batteries, unit equipment or on-site generator permitted	Direct	Coordinate 90-minute basis with electrical schedule	Verified
1008.3.5	Initial emergency level	Average at least 11 lux , minimum at any point 1 lux at floor level along path	Emergency power starts	No project-specific reduction	Direct	Verify average and point minimum by calculation	Verified
1008.3.5	End-of-duration level	May decline to average 6 lux , point minimum 0.6 lux at end of duration	At 90-minute endpoint	Applies only at end of emergency-lighting duration	Direct	Include depreciation/end-duration check	Verified
1013.1	Exit-sign spacing	No point in exit-access corridor or exit passageway more than 30 m or listed viewing distance, whichever is less, from visible sign	Direction to exit not readily apparent	Signs not required inside individual R-2 sleeping/dwelling units; one-exit rooms and obvious approved main exterior exits also listed	Direct	Prepare sign visibility/spacing plan outside units	Verified
1013.6.1–1013.6.3	Exit-sign graphics and power	Letters at least 150 mm high with 20 mm strokes; “EXIT” letter width at least 50 mm except I, spacing at least 10 mm ; externally lit face at least 55 lux ; backup at least 90 minutes	Exit/directional sign required	Approved self-powered sign may satisfy stated 90-minute exception	Direct	Coordinate sign schedule and photometrics	Verified
1023.9–1023.9.1	Stair identification sign	Required at each landing when stair/ramp connects more than 3 stories ; bottom at least 1500 mm above landing; sign at least 450 × 300 mm , stair ID 40 mm , floor number 125 mm , other text 25 mm	High-rise interior exit stair/ramp	Tactile sign requirements are separately cross-referenced	Direct	Issue a consistent stair-sign family and location detail	Verified

10. Doors, gates and turnstiles

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1010.1.1	Door clear opening	Capacity-sized and at least 800 mm clear width ; minimum clear height 2000 mm	Required egress door	R-2 internal nonaccessible-unit exceptions exist; Type B user-passage doors remain 800 mm	Direct	Schedule clear opening, not nominal leaf size	Verified
1010.1.1	Swing-door measurement	Measure between door face and stop with door at 90 degrees ; one leaf of a pair without mullion must provide 800 mm clear	Swinging/pair doors	Listed special doors have separate provisions	Direct	Dimension clear width at 90 degrees in door details	Verified
1010.1.2.1	Direction of swing	Door swings in egress direction for room/area load 50 or more , or Group H	Threshold met	Unit entrance below threshold is not automatically required by this clause to swing out	Conditional	Link each door swing to served room load	Verified
1010.1.3	Door operating forces	Push/pull unlatching maximum 67 N ; rotational unlatching maximum 3.15 N-m ; manually operated interior swinging non-fire-rated egress door opening maximum 22 N ; other, sliding, folding and fire-rated doors maximum 133 N to set in motion and 67 N to full-open	Manually operated egress door	Power-operated doors are addressed separately; forces are applied at latch side under 1010.1.3.1	Direct	Put force criteria in hardware specification and commissioning checks	Verified
1010.1.5	Door landing	Landing length in travel direction at least 1100 mm ; fully open door may reduce required dimension by no more than 175 mm	Egress door landing	Within individual R-2 units, landing length need not exceed 900 mm	Direct	Check both open and closed door positions	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1010.1.6	Threshold — base	Sliding doors serving dwelling units maximum 20 mm ; other doors maximum 12.5 mm ; changes over 6 mm bevel no steeper than 1:2 (50%)	Doorway threshold	Exceptions below are narrowly scoped	Direct	Use the base limits for required egress and accessible doors	Verified
1010.1.6 Ex. 1	R-2 exterior threshold	Sliding or side-hinged exterior threshold may be up to 200 mm only where the door is not part of required egress, not part of a Chapter 11 accessible route, and not part of an Accessible, Type A or Type B unit	Group R-2 or R-3 exterior door meeting all three conditions	Failure of any condition returns the door to the base limits	Conditional	Identify the nonrequired, nonaccessible door and unit type before using 200 mm	Verified
1010.1.6 Ex. 2	Type B exterior-side threshold	Where 1010.1.4 Ex. 5 permits a 100 mm exterior level change, exterior-side threshold maximum is 120 mm above deck/patio/balcony for sliding doors and 115 mm for other doors	Exterior door of Type B unit under the cited level-change exception	These are exterior-side measurements; base interior-side limits and beveling still apply	Conditional	Detail both interior and exterior threshold datums	Verified
Unnumbered — commentary indicates 1010.1.7	Doors in series / vestibule	Space between two doors in series is 1200 mm minimum plus the width of a door swinging into the space ; doors swing in the same direction or away from the intervening space	Successive doors/vestibule on an egress path	Horizontal sliding power doors: 1200 mm minimum; R-2/R-3 unit storm/screen doors need not be spaced 1200 mm; doors within R-2/R-3 dwelling units are excepted other than within Type A dwelling units	Conditional	Dimension vestibules beyond the full door swing; apply unit exceptions only at their exact scope	Verify source
1010.2.1	Unlatching	Not more than one motion in one linear/rotational direction to release latching/locking devices	Egress side of doors	Group R unit security devices are a listed exception under 1010.2.4 Item 5	Direct	Coordinate hardware sets for single-action egress	Verified
1010.2.3	Hardware height	Operating devices 850–1200 mm above finished floor	Egress doors/gates	Pool/spa barrier latch exception up to 1350 mm under stated conditions	Direct	Put range in hardware specification	Verified
1010.2.4 Item 5	R-unit security	Group R dwelling/sleeping-unit doors serving load 10 or less may have night latch, dead bolt or security chain if openable inside without key/tool	Individual unit	Does not waive one-motion/safe-egress intent beyond stated permission	Conditional	Confirm unit occupant load and free-egress operation	Verified
1010.2.9	Panic hardware	Group A/E rooms/spaces with load 50 or more require panic/fire-exit hardware; Group H also listed	Assembly amenity or mixed-use trigger	Main-exit and electrical locking exceptions are listed	Conditional	Check amenity and podium doors by actual occupancy/load	Verified
1010.2.11 Item 2	Hardware-released electric lock	Door-mounted release hardware is operable with one hand and complies with 1010.2.1	Door-hardware release electric locking system, except Group H	All listed release and power-loss criteria remain cumulative	Conditional	Coordinate the hardware set and single-hand operation with the electrical lock release	Verified
1010.2.12	Sensor-released lock	Manual release at 1000–1200 mm high and within 1500 mm of secured doors; direct power interruption; unlock at least 30 seconds	Sensor-released electrically locked egress door	Fire alarm, sprinkler/detection and power-loss release conditions also required	Conditional	Coordinate access control with fire alarm and emergency lighting	Verified
1010.2.13 / 1010.2.13.1	Delayed egress	General path maximum one delayed-egress system ; irreversible release within 15 seconds after no more than 3 seconds applied effort; approved delay may be 30 seconds	Delayed-egress locking system serving Group R or another listed occupancy with required sprinkler/detection prerequisite	Institutional two-device exceptions are NOT_TYPICAL and capped at 30 seconds combined	Conditional	Treat delayed egress as a fire-alarm, access-control, emergency-lighting and signage package; obtain approval for any 30-second branch	Verified
1010.2.13.1 Item 6	Delayed-egress sign	Sign above and within 300 mm of exit hardware; PUSH/PULL legend states opening in 15 [30] seconds according to door swing	Delayed-egress door	Group I sign omission is a separate clinical exception	Conditional	Put exact legend, location and approved delay on the door/hardware schedule	Verified
1010.3.1	Revolving door	Breakout aggregate width at least 900 mm ; side-hinged 1010.1 door in same wall within 3000 mm ; revolving door not on accessible route	Revolving entrance proposed	Egress credit limited to 50% and 50 persons per door under 1010.3.1.1	Conditional	Prefer compliant side-hinged egress doors as the dependable route	Verified
1010.5.1	Conventional turnstile	Credit maximum 50 persons and no more than 50% required capacity; device max 990 mm high; clear width 420 mm up to 990 mm high and 550 mm above	Turnstile used for egress credit	Must free-turn on power loss/manual release	Conditional	Keep at least half capacity independent of turnstiles	Verified
1010.5.1.1	Accessible-route turnstile	Clear width at least 900 mm at/below 850 mm high and at least 800 mm between 850–2000 mm ; mechanism is not revolving	Turnstile lies on an accessible route	Accessibility geometry and approach remain separately controlled	Conditional	Provide a compliant nonrevolving accessible lane and dimension both height branches	Verified
1010.5.2	Security access turnstile	Lane minimum 550 mm ; lane under 800 mm credited maximum 50 persons ; lane 800 mm or more sized under Section 1005	Approved supervised 903.3.1.1 sprinklered building and listed release conditions	Requires automatic unobstructed opening on specified events	Conditional	Use only with NFPA 13 branch and coordinated fail-safe release	Verified
1010.5.4	Turnstile companion door	For served load over 300 , side-hinged compliant door within 15 m	Nonportable turnstile	Exception for compliant 1010.5.2 security turnstiles	Conditional	Include companion door in lobby planning if trigger is met	Verified

Door-arrangement numbering hold point: the supplied Chapter 10 text omits the section number before “Door arrangement”; its commentary identifies **1010.1.7**. The mandatory values are retained under the citation **Unnumbered — commentary indicates 1010.1.7**, and the published section number remains a verification hold point.

11. Stairs

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1011.2	Minimum width	Capacity-sized under 1005 and at least 1100 mm	General stair	Stair serving fewer than 50 occupants may be 900 mm ; accessible-egress clear width is separate	Direct	Use the larger of capacity, 1100 mm and accessible-egress requirement	Verified
1011.3	Headroom	At least 2100 mm vertically from nosing line, continuous to one tread beyond bottom riser	General stair	Spiral stair 2000 mm ; limited R-2 in-unit projection exception 120 mm	Direct	Check soffits, landings and underside of flights in section	Verified
1011.5.2	Public/common riser and tread	Riser 100–175 mm ; rectangular tread at least 275 mm	Common/public tower stairs	Assembly stepped aisles and listed special stairs separate	Direct	Apply to exit stairs and common circulation stairs	Verified
1011.5.2 Ex. 3	Inside individual R-2 dwelling	Riser maximum 200 mm ; tread and winder walkline at least 250 mm ; minimum winder depth 150 mm ; solid-riser nosing 20–32 mm where tread below 275 mm	Stair wholly within individual dwelling unit	Not for common/exit stair	Conditional	Label in-unit stairs distinctly so exception is not propagated to core stairs	Verified
1011.5.4	Uniformity	Largest-smallest riser or tread variation maximum 10 mm in a flight	All flights	Listed stepped-aisle and consistent-winder branches	Direct	Add construction tolerance requirement	Verified
1011.6	Landing depth and door	Landing depth equals stair width or 1200 mm , whichever is less; fully open door projects no more than 175 mm ; open door cannot reduce landing below one-half required width	Stair landing	Listed transition/curved landing exceptions	Direct	Overlay door swings on clear landing diagrams	Verified
1011.7.1	Stair walking-surface slope	Maximum 1:48 (2%) in any direction	Treads and landings	Opening-size exceptions for industrial uses	Direct	Coordinate drainage without exceeding cross slope	Verified
1011.8	Maximum flight rise	3600 mm maximum vertical rise between landings	Stair flight	Technical-production spiral exception	Direct	Insert intermediate landing where floor-to-floor geometry requires	Verified
1011.12	Roof stair	In buildings 4 or more stories , one stair extends to roof unless roof slope exceeds 4:12 (33%)	Tower roof	Unoccupied-roof access alternatives listed except where 1011.12.1 requires otherwise	Conditional	Confirm occupied/unoccupied roof and coordinate one core to roof	Verified
1011.12.2	Unoccupied-roof hatch	Hatch permitted at least 1.5 m² , minimum dimension 600 mm	Building without occupied roof	Applies only under stated roof-access exception	Conditional	Coordinate hatch, ladder/device and roof safety	Verified
1011.13	Roof-edge guard near hatch	Guard roof hatch/access or roof edge where opening is within 3000 mm of roof edge	Roof access near edge	Guard must satisfy 1015	Conditional	Keep hatch farther from edge or detail compliant guard	Verified

12. Ramps

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1012.2	Means-of-egress ramp slope	Maximum 1:12 (8.3%)	Ramp forms part of means of egress	Other pedestrian ramps may be 1:8 (12.5%) , but accessible route may be more restrictive outbound	Conditional	Use 1:12 for egress/accessible planning unless verified otherwise	Verified
1012.3	Ramp cross slope	Maximum 1:48 (2%)	All ramps	No numeric branch stated	Conditional	Coordinate drainage falls	Verified
1012.4	Rise per run	Maximum 750 mm	Each ramp run	No numeric branch stated	Conditional	Set landing frequency from rise	Verified
1012.5.1	Ramp clear width	At least 900 mm between handrails/projections and sized for capacity	Egress ramp	Accessible route and served load may require more	Conditional	Check clear width after handrails	Verified
1012.5.2	Ramp headroom	At least 2000 mm over runs and intermediate landings	Egress ramp	No stated numeric exception	Conditional	Coordinate structure/services in section	Verified
1012.6.3–1012.6.4	Ramp landings	Length at least 1500 mm ; direction-change landing 1500 × 1500 mm	Ramp landing	Nonaccessible R-2 individual units may use 900 mm landing / 900 × 900 mm turn; nonaccessible ramp length need not exceed 1200 mm	Conditional	Apply accessibility/unit classification before selecting exception	Verified
1012.8	Ramp handrails	Both sides where rise exceeds 150 mm	Ramp run	No general numeric exception in record	Conditional	Detail continuous handrails and extensions	Verified
1012.10.1–1012.10.2	Edge protection	Curb at least 100 mm , or qualifying barrier; extended floor/ground at least 300 mm beyond inside face of handrail	Ramp/landing edge protection required	Listed flared-side/adjoining-run/minor-drop exceptions	Conditional	Select and detail one compliant edge-protection method	Verified

13. Handrails and guards

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1014.2	Handrail height	Uniform 850–950 mm above stair nosings or ramp surface	Required handrails	Alternating tread/ship ladder 750–850 mm ; fitting transitions listed	Direct	Dimension from correct datum, not finished landing	Verified
1014.3.1	Type I graspability	Circular outside diameter 32–50 mm ; noncircular perimeter 100–160 mm , max cross-section 60 mm , min 25 mm	Type I handrail	Edge radius at least 0.25 mm	Direct	Put profile limits in metalwork/joinery details	Verified
1014.6	Extensions	Ramp extensions 300 mm horizontally beyond top/bottom; stair top 300 mm , bottom slopes one tread depth	Handrails not continuous between runs/flights	Nonaccessible dwelling-unit and assembly-aisle exceptions	Direct	Resolve extensions against doors and circulation	Verified
1014.7	Wall clearance	At least 38 mm clear between handrail and wall/surface	Wall-mounted handrail	No numeric branch stated	Direct	Coordinate brackets and finishes	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1014.8	Projection	Handrail projections into required aisle/stair/ramp width maximum 115 mm each side; accessible ramp clear between rails at least 900 mm	Handrails project into egress width	Intermediate-pair rule applies where gap over 150 mm	Direct	Calculate capacity on net clear width	Verified
1014.9	Intermediate handrails	All portions of stair minimum width/required capacity within 750 mm of a handrail	Wide stair	No stated numeric exception	Conditional	Add intermediate rails to wide stairs	Verified
1015.2	Guard trigger	Guard open-sided walking surfaces where drop exceeds 750 mm at a point within 900 mm horizontally of edge	Mezzanine, balcony, stair, ramp, landing, roof edge	Listed stage/loading/service exceptions	Direct	Survey every edge and adjacent level	Verified
1015.3	General guard height	At least 1100 mm	Required guards	R-2 in-unit stair guards may be 850 mm ; top-as-handrail 850–950 mm ; low-rise in-unit 900 mm exception does not describe this high-rise tower generally	Direct	Use 1100 mm for common areas and document any in-unit exception	Verified
1015.4	Guard openings	General guard rejects 100 mm sphere; in R-2 unit open stair, 110 mm sphere; triangular stair opening 150 mm	Required guard	Other listed industrial/assembly branches	Direct	Specify opening limits by location	Verified
1015.6–1015.7	Roof service/access guards	Guards where service component or hatch is within 3000 mm / 3 m of edge and edge drop exceeds 750 mm ; service guard extends 750 mm beyond component/hatch	Roof equipment/access near edge	Fall-arrest/restraint anchorage exception under stated standard	Conditional	Coordinate roof plant zones, access path and fall protection	Verified
1015.8	R-2 operable-window trigger	In Group R-2/R-3 buildings including dwelling units, comply with one of four controls where bottom of clear operable opening is less than 900 mm above interior floor and exterior drop is more than 1800 mm to grade/surface below	Both sill/drop conditions occur	Raising bottom of clear opening to at least 900 mm avoids this trigger; an EERO must also satisfy 1031	Direct	Audit every operable residential window against interior sill and exterior drop	Verified
1015.8 Items 1–3	Window fall-prevention branches	(1) Where top of sill is more than 23 m above exterior grade/surface, provide ASTM F2006 device; or (2) largest open position rejects a 100 mm-diameter sphere ; or (3) provide ASTM F2090 fall-prevention device	Triggered R-2/R-3 operable window	ASTM F2006 branch is expressly tied to the more-than-23 m top-of-sill condition; select one listed branch	Direct	Coordinate façade hardware and identify any window also serving as EERO	Verified
1015.8 Item 4 / 1015.8.1	Window opening-control device	Opening-control device complies with ASTM F2090 and, after release for full opening, does not reduce net clear opening below Section 1031.3.1	Triggered window uses opening-control-device branch	Section 1031.3.1 criterion matters where the window is an EERO	Conditional	Obtain listed device data and verify released EERO clear area where applicable	Verified

14. Corridors, aisles and exterior egress balconies

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1016.2	Intervening spaces	Exit access cannot pass through a lockable room; dwelling/sleeping-area egress cannot lead through another sleeping area, toilet or bathroom; egress generally cannot pass through kitchens/storage/closets	Exit route passes through another space	Kitchen serving adjoining rooms in same dwelling/sleeping unit is a listed exception	Direct	Trace unit and shared routes against room locking and use	Verified
1016.2	Elevator lobby route	If two or more exits are required, at least one required exit must be accessible without travel through the enclosed elevator lobby required by Section 3006	Enclosed elevator lobby lies on exit access	Lobby protection need not extend to exit unless another section requires direct access	Conditional	Preserve one independent route around lobby	Verified
1016.2.1	Multiple tenants/units	Each tenant space, dwelling unit and sleeping unit gets access to required exits without passing through adjacent tenant/unit	More than one tenant/unit on floor	Smaller-tenant exception is less than 10% of larger adjoining space and has other conditions	Direct	Keep residential units independently connected to common egress	Verified
1018.3 / 1018.5	Nonassembly aisles	Nonpublic aisle serving fewer than 50 and not required accessible may be 700 mm	B/M or other nonassembly work/stock aisle	Public/accessible aisle must satisfy applicable width/capacity	Conditional	Apply only to genuinely nonpublic support aisles	Verified
Table 1020.2	R corridor rating	For Group R corridor serving more than 10 , OCR reads unsprinklered Not Permitted and sprinklered “ 0.5/1 hours ”; rating/footnote attachment unresolved	Common R corridor	Table recognizes 903.3.1.1 or 903.3.1.2 where allowed; in-unit R corridor rating exception exists	External verification	Obtain published table/footnote before selecting corridor rating	Verify source
Table 1020.2	Corridor width	OCR table contains 1100 mm general and 900 mm small-load/within-unit values, but row attachment is unresolved	Corridor width selection	Capacity under Section 1005 can govern above table minimum	External verification	Verify published table before issue; provisionally coordinate common corridor no narrower than capacity and verified minimum	Verify source
1021.3	Balcony openness	Long side at least 50% open , distributed to minimize smoke/toxic-gas accumulation	Exterior egress balcony	Wall separation and opening protection also apply	Conditional	Demonstrate permanent openness, not operable glazing	Verified
1021.4	Balcony location	Fire separation distance at least 3000 mm from lot line, other building portions and same-lot buildings unless protected as stated	Exterior egress balcony	Section 705 protection branch	Conditional	Coordinate façade returns and adjacent masses	Verified

Dead-end corridor limits (base **6 m**; **15 m** only with Section 903.3.1.1 sprinklers) and the **30 m** exterior-balcony travel increase are tabulated once in Section 7.

15. Interior exit stairs/ramps, exit passageways and horizontal exits

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1023.1	Exit enclosure continuity	Interior exit stair/ramp enclosed and leads directly outside or extends outside through 1024 exit passageway; used only for egress and circulation	Interior exit stair/ramp	Limited discharge alternatives under 1028.2	Direct	Keep storage, plant and non-egress uses out of enclosure	Verified
1023.2	Enclosure rating	2 hours where connecting 4 stories or more ; 1 hour where fewer; basements count, mezzanines do not	Interior exit enclosure	Rating not less than penetrated floor and need not exceed 2 hours; listed special exceptions	Direct	Tower core should be coordinated to 2-hour branch, subject to full code review	Verified
1023.4	Openings	Openings limited to required exit access and egress; elevators shall not open into interior exit stairways/ramps	Exit enclosure	Unprotected exterior openings handled separately	Direct	Eliminate service-room and elevator openings into stairs	Verified
1023.7	Exposed exterior wall	At angle below 180° , building wall within 3000 mm requires at least 1 hour ; protection extends to 3000 mm above top landing/roof line; openings at least 3/4 hour	Stair exterior wall exposed by building return	Applies to affected exposure zone	Conditional	Check façade returns around discharge and roof	Verified
1023.10 / 1023.9.1 Items 4–6	Elevator-lobby identification	At an interior-exit-stair landing where two or more doors lead to the floor , any door directly accessing an enclosed elevator lobby is marked “ Elevator Lobby ” on or directly adjacent to the door; lettering/numbers at least 25 mm , nonglare and contrasting	Two-or-more-door stair landing with direct enclosed-lobby door	If the building/sign is subject to Section 1025, use the same material required by 1025.4; R-2 alone is not a listed 1025.1 group	Conditional	Add the lobby-door sign to stair-core door/sign schedules without importing Section 403 elevator assumptions	Verified
1024.2	Exit-passageway width	Capacity-sized; minimum 1100 mm , or 900 mm when serving fewer than 50 occupants	Exit passageway used	Permitted encroachments under 1005.7	Conditional	Size for cumulative stair load, not only local floor	Verified
1024.3	Exit-passageway rating	At least 1 hour and not less than connected stair/ramp requirement	Exit passageway extends protected exit	Connected 2-hour tower stair therefore controls at not less than that rating	Conditional	Match passageway protection to connected enclosure	Verified
1026.1	Horizontal-exit share	Cannot be sole exit; with multiple exits, no more than one-half total number or width/capacity may be horizontal exits	Horizontal exit proposed	I-2/I-3 exceptions not typical for R-2	Conditional	Retain independent vertical/outdoor exits for remaining required share	Verified
1026.2	Horizontal-exit separation	Minimum 2-hour separation, extending vertically unless floors are at least 2 hours with no unprotected openings	Horizontal exit proposed	Pedestrian-walkway exception only under stated conditions	Conditional	Align barrier continuity through tower/podium	Verified
1026.4.1	Refuge capacity	Net 0.28 m²/person accommodated	Horizontal-exit refuge area	Separate from 1009 wheelchair-space geometry	Conditional	Calculate both total refuge area and accessible spaces	Verified
1026.4.2	Refuge exit	At least one refuge-area exit leads directly outside or to interior exit stair/ramp	Horizontal-exit refuge	Listed adjoining-compartment exception	Conditional	Show onward route without return through fire-origin compartment	Verified

16. Exit discharge and egress courts

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1028.1 / 1028.5	Discharge destination	Exits discharge directly outside; exit discharge provides direct, unobstructed access to public way	Every exit	Limited 1028.2 discharge-through-area/vestibule/horizontal-exit branches	Direct	Trace every stair/exit independently to public way	Verified
1028.2 Ex. 1	Discharge through level area	No more than 50% of number and width/capacity of interior exit stairs/ramps may egress through discharge-level areas	Lobby/atrium discharge proposed	Requires visible unobstructed exterior door and separation below; egress path is sprinkler-protected, and portions with access to it are either sprinklered throughout under 903.3.1.1 or 903.3.1.2 or separated as an interior-exit enclosure	Conditional	Keep at least half direct and document the NFPA 13 or NFPA 13R discharge-lobby protection branch	Verified
1028.2 Ex. 1.4	Stair termination separation	Exit-access stair/ramp termination and interior-exit discharge door separated by at least 9 m or one-fourth building diagonal, whichever is less	Both terminate at same discharge level	Straight-line measurement	Conditional	Dimension conflict separation in lobby plan	Verified
1028.2 Ex. 2	Vestibule discharge	No more than 50% of number and width/capacity may discharge through vestibule	Vestibule alternative	Depth from exterior max 3000 mm , length max 9 m ; protected and used only for egress	Conditional	Detail enclosure, dimensions and direct outside door	Verified
1028.2	Combined exceptions	Combined use of area and vestibule exceptions cannot exceed 50% of required exits and width/capacity	Both exceptions used	Aggregate cap, not 50% each	Conditional	Include combined schedule	Verified
1028.5	Safe dispersal	If public-way access cannot be provided: at least 0.46 m²/person , on same lot at least 15 m from building, permanently maintained/identified, with safe unobstructed path	Safe-dispersal alternative	Only when direct public-way access cannot be provided	Conditional	Coordinate site/fire-appliance/landscape plans and capacity	Verified
1029.2	Egress-court width/headroom	Capacity-sized; minimum 1100 mm ; unobstructed to 2.1 m high	Egress court forms discharge	R-3/U 900 mm exception does not apply to R-2 tower	Conditional	Protect net width from gates, planting and services	Verified
1029.3	Narrow egress court	Where court is under 3000 mm wide, walls at least 1 hour for 3000 mm above court floor; openings at least 3/4 hour	Narrow court	Exceptions for load under 10 and R-3, not general R-2	Conditional	Coordinate wall/opening ratings along court	Verified

17. Assembly amenity conditions

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
Table 1004.5	Unfixed amenity seating/load	Standing 0.46 m² net/person ; chairs only 0.65 m² net/person ; tables/chairs 1.4 m² net/person	Assembly amenity without fixed seats	Select by actual configuration; pool/deck factors unresolved	Conditional	Calculate each operating/event layout	Verified
1004.9	Occupant-load posting	Every assembly room/space posts approved load conspicuously near main exit/access doorway in permanent, maintained form	Space is assembly occupancy	Intended configurations must be represented	Conditional	Add placard to signage schedule	Verified
1030.2	Main exit over 300	If assembly load is greater than 300 and a main exit is provided, main exit accommodates at least one-half load and at least total capacity leading to it	Large assembly amenity	If no well-defined main exit/multiple main exits, perimeter distribution may provide 100% capacity	Conditional	Test ballroom/cinema/club layouts if load crosses 300	Verified
1030.2	Group A frontage	Group A main exit fronts a street or at least 3000 mm unoccupied space adjoining street/public way	Building/space classified Group A and main exit used	Occupancy classification is outbound from Chapter 10	Conditional	Confirm whether amenity is Group A and coordinate frontage	Verified
1030.3	Other exits over 300	In addition to main exit, each level with load over 300 gets additional egress capacity for at least one-half total load served, complying with 1007.1	Large assembly with main exit	Distributed-main-exit alternative separately addressed	Conditional	Provide remote additional exits and capacity split	Verified
1030.5	Balcony/gallery	Seating capacity 50 or more requires at least two means of egress , one from each side	Interior assembly balcony/gallery/press box	Under 50 follows other applicable rules	Conditional	Avoid single-ended amenity mezzanine seating at trigger	Verified
1030.8	Assembly common path	Base maximum 9 m from seat to choice of two routes; area serving fewer than 50 may use 23 m	Assembly seating	Smoke-protected/open-air branch 15 m	Conditional	Measure from worst seat under every furniture plan	Verified
1030.9.1	Assembly aisle widths	Stepped, seating both sides 1200 mm ; one side 900 mm ; level/ramped both sides 1100 mm ; one side 900 mm	Assembly seating aisle	Listed reductions to 900/750/600 mm depend on seat count, accessibility and handrail arrangement	Conditional	Size from served seats and accessible route	Verified
1030.13.1	Seating at tables datum	Required clear width measured to line 480 mm from and parallel to table/counter edge	Loose seating at tables/counters	Fixed-seat measurement from back of seat	Conditional	Draw chair occupancy envelope in egress plans	Verified
1030.13.1.1	Table aisle access-way	At least 300 mm plus 12.5 mm per additional 300 mm/fraction beyond 3600 mm length	Seating at tables/counters	Portion not over 1800 mm serving max 4 persons excepted	Conditional	Check deep banquettes and private dining layouts	Verified
1030.13.1.2	Table access travel	Maximum 9 m from any seat to choice of two or more separate-exit paths	Seating at tables	No numeric branch stated	Conditional	Test farthest chair, not table edge	Verified
1030.14.1	Ramped assembly aisle	Over 1:20 (5%) is a ramped aisle; accessible route max 1:12 (8%) ; other ramped aisle max 1:8 (12.5%)	Sloped amenity seating aisle	Cross slope max 1:48 (2%)	Conditional	Coordinate sightlines with accessible slope	Verified
1030.14.2	Stepped assembly aisle	Slope over 1:8 (12.5%) uses full-width risers/treads; tread at least 275 mm ; riser generally 100–200 mm	Stepped seating aisle	Sightline exception permits riser to 225 mm with stated marking/tolerance controls	Conditional	Use dedicated stepped-aisle detail, not ordinary stair detail	Verified
1030.16	Assembly aisle handrails	Ramped aisle over 1:15 (6.7%) and stepped aisle require handrails; wide one-sided stepped aisle 1850 mm or more requires two; one rail within 750 mm	Assembly aisles	Listed seating/guard and extension exceptions	Conditional	Coordinate mid-aisle rails with seat access	Verified

18. Emergency escape and rescue openings

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1031.2 / Table 1006.3.4	R-2 scope	Group R-2 is subject to 1031.2 only when located in a story with one exit or access to one exit as permitted by Tables 1006.3.4(1)/(2)	Project relies on a permitted single-exit R-2 story	General multiexit R-2 floors are not independently brought into 1031.2 by the R-2 listing	Conditional	Keep EEROs within the complete, source-verified single-exit package	Verified
1031.2	Location rule by listed occupancy	Within the listed occupancy scope, basements and sleeping rooms below the fourth story above grade plane require at least one EERO; each basement sleeping room requires one and adjoining basement areas do not	R-3/R-4 generally; R-2 only within the single-exit scope above	Opening leads directly to public way or a yard/court opening to public way; listed exceptions follow	Conditional	Do not apply the general R-3/R-4 trigger to ordinary multiexit R-2 tower floors	Verified
1031.2 Ex. 5	Sprinklered basement sleeping rooms	In an individual R-2/R-3 dwelling or sleeping unit, basement sleeping-room EEROs may be omitted where building is throughout sprinklered under 903.3.1.1, .1.2 or .1.3 and basement has either one means of egress plus one EERO , or two means of egress	R-3 generally; R-2 only where 1031.2 already applies through the permitted single-exit-story scope	Exception changes basement sleeping-room distribution; it does not expand 1031.2 to general multiexit R-2	Conditional	Confirm the parent 1031.2 scope before using Exception 5	Verified
1031.3.1	Net clear area	Minimum 0.53 m² ; grade-floor opening minimum 0.46 m²	EERO required	Grade-floor exception only	Conditional	Schedule net clear area in open position	Verified
1031.3.2	Minimum dimensions	Clear height at least 600 mm and clear width at least 500 mm	EERO required	Both dimensions apply alongside area	Conditional	Verify proprietary window free opening, not frame size	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1031.3.3	Sill height	Bottom of clear opening maximum 1100 mm above floor	EERO required	No numeric branch stated	Conditional	Coordinate furniture and façade module	Verified
1031.5.1	Area well	Minimum horizontal area 0.84 m² , projection and width at least 900 mm	EERO opens into area well	Ladder/steps may encroach maximum 150 mm	Conditional	Coordinate structure, drainage and usable clear opening	Verified
1031.5.2	Deep well ladder/steps	Well deeper than 1100 mm requires permanently fixed ladder or steps	Deep area well	Must not obstruct opening operation	Conditional	Detail access and opening swing together	Verified
1031.5.2.1–.2	Well ladder/step geometry	Ladder inside width at least 300 mm , projection at least 75 mm , vertical spacing max 450 mm ; steps inside width at least 350 mm , tread over 125 mm , riser max 450 mm	Ladder/steps provided	Use the applicable device branch	Conditional	Include complete dimensional detail	Verified

19. High-rise and outbound dependencies

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1023.12	Smokeproof enclosure dependency	“Where required by Section 403.5.4, 405.7.2 or 412.2.2.1,” interior exit stairs/ramps are smokeproof enclosures under 909.20	High-rise statement points toward 403.5.4 review	Chapter 10 does not fully define the high-rise trigger, exceptions or 909.20 system	External verification	Verify Section 403 applicability/exceptions and engineer the 909.20 system; do not claim Chapter 10 alone proves smokeproof stairs	Verified
1009.2.1 / 1009.4	Accessible-egress elevator	Four-or-more-storey trigger is in Chapter 10; elevator must comply with 1009.4	Required accessible floor/roof at trigger	1009.4 then cross-references elevator systems/standby power and other requirements	External verification	Coordinate accessibility, vertical transport, fire alarm and fire strategy	Verified
Section 403 / Ch. 30	High-rise/fire-service/occupant elevators	Chapter 10 commentary/cross-references identify high-rise elevator interfaces, but Chapter 10 does not fully define Section 403 elevator triggers or values	High-rise tower	Requirements and exceptions lie outside Chapter 10	External verification	Do not claim high-rise elevator quantities/features from this matrix; verify Sections 403, 3007 and 3008 as applicable	Verified
1025.1	Luminous egress path markings	Listed high-rise groups are A, B, E, I-1, M and R-1	High-rise building in a listed group	R-2 is not included in the listed groups; mixed-use listed occupancy portions may still require analysis	Not typical	Do not impose Section 1025 solely because the tower is R-2 high-rise; review any listed podium/mixed use separately	Verified
1005.3.1 Ex. 1	Reduced stair factor systems	5.08 mm/person only with 903.3.1.1 or .1.2 sprinklers plus 907.5.2.2 EVACS	Capacity reduction proposed	High-rise sprinkler/EVACS requirements themselves are outside this clause	External verification	Keep 7.6 mm/person until fire engineer confirms both systems and AHJ acceptance	Verified
Table 1017.2	Travel-distance dependency	Group R row OCR reads “ 60” / 75” ”, unresolved	Travel design	Published table/footnotes and sprinkler basis need verification	External verification	Resolve before schematic egress freeze	Verify source
Table 1020.2	Corridor dependency	Group R sprinklered rating OCR reads “ 0.5/1 hours ”, unresolved	Common R corridor serving load over 10	Published table and footnote attachment required	External verification	Do not release partition rating from this extract	Verify source
1002.1–1002.2	Operations and plans	Means of egress maintenance and required fire-safety/evacuation plans are directed to SBC 801	Occupancy/operation	Chapter 10 does not define the full operational plan	External verification	Integrate assisted evacuation, phased strategy and management procedures into approved fire strategy/SBC 801 plan	Verified
Chapter 11 / ICC A117.1	Accessibility geometry	Chapter 10 establishes accessible-egress scoping but cross-references technical accessibility requirements	Accessible routes, doors, signs, ramps and communication	Exact technical geometry outside the Chapter 10 source-only matrix	External verification	Complete separate SBC accessibility review	Verified
Chapters 7 and 9 / SBC 501	Passive and active fire protection	Chapter 10 relies on fire barriers, opening protectives, sprinklers, alarm and smoke-control provisions elsewhere	All protected egress systems	Ratings and system design are not fully specified by Chapter 10 alone	External verification	Align architectural egress matrix with fire engineer’s code basis and SCD NOC package	Verified
SCD / SBPS	Authority acceptance	No Chapter 10 value constitutes authority approval	Riyadh permit/NOC pathway	AHJ identity, submission stage and comments unconfirmed	External verification	Qualified local consultant to confirm authority pathway and close decision register before issue for approval	Verified

20. Project-use controls

1. Use the **Verified** rows for initial dimensional coordination only after the row trigger and branch are confirmed.
2. Treat every **Verify source** row as a design hold point; no affected value is to be placed in issued-for-approval drawings without a published-source check.
3. Maintain separate NFPA 13 and NFPA 13R life-safety diagrams until the fire strategy selects one. In particular, the Section 1020.5 **15 m** R-2 dead-end allowance is NFPA 13 only.
4. Calculate occupant load by actual room function, then test in sequence: room/space exit-access doors, story exits, remoteness, capacity, loss of one exit, travel/common path, accessible egress, discharge and public-way continuity.
5. Record all fire-strategy and authority decisions in the project Golden Thread; this matrix is not evidence of SCD NOC or stamped compliance.

21. Coverage summary

Internal inventory of the attached Chapter 10 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Total independently checkable numeric records:** 645
- **Verified:** 557
- **Verify source:** 88
- **Numeric records in Sections 1001 and 1002:** 0

Counts by top-level section

Top-level section	Records
1001	0
1002	0
1003	15
1004	47
1005	11
1006	49
1007	7
1008	12
1009	15
1010	116
1011	55
1012	17
1013	10
1014	18
1015	24
1016	5
1017	16
1018	7
1019	1
1020	21
1021	3
1022	1
1023	14
1024	6
1025	20
1026	8
1027	10
1028	7
1029	5
1030	109
1031	16

Appended-table coverage

Appended table	Records	Verify source records
Table 1004.5	38	6
Table 1006.2.1	14	14
Table 1006.3.3	3	0
Table 1006.3.4(1)	1	1
Table 1006.3.4(2)	8	8
Table 1010.3.1(1)	5	0
Table 1010.3.1(2)	12	12
Table 1017.2	11	11
Table 1020.2 — corridor rating	7	7
Table 1020.2 — minimum width	7	7
Table 1030.6.2	5	0
Table 1030.13.2.1	8	8

22. Unresolved-source register

Hold points for the 88 **Verify source** inventory records. Counts are record counts, not distinct numeric values. No value in this register is a design-release figure.

Affected table / clause	Why unverified	Control note
Table 1004.5	6 flattened/unresolved rows	Verify published table; empty values not adopted
Section 1005.3.2	Exception 1 OCR corrupted	No reduced non-stair factor inferred
Table 1006.2.1	14 flattened rows; footnote attachment unresolved	Verify published table before one-door / common-path use
Section 1006.2.2.2	Page-split continuation	Verify published clause
Table 1006.3.4(1)	Flattened single-exit dwelling-unit row	Verify published table
Table 1006.3.4(2)	8 flattened single-exit rows	Verify published table; do not transpose dwelling/sleeping branches
Section 1008.2.2	Grammatical fragment in mandatory OCR	Verify published clause
Section 1009.2.1	Duplicated wording in Exception 2	Base four-storey elevator trigger is used; exception wording needs published check
Unnumbered — commentary indicates 1010.1.7	Source omits section number	Door-arrangement values retained under this citation
Table 1010.3.1(2)	12 concatenated diameter/speed cells	Verify published table
Section 1011.2	Exception 3 crosses pages	Verify published clause
Table 1017.2	Damaged header / degree-symbol cells (60° / 75°)	Do not treat as metres
Table 1020.2 — corridor rating	Duplicate OCR table; Group R 0.5/1 unresolved	Do not release partition rating from this extract
Table 1020.2 — minimum width	Duplicate OCR table; row/footnote attachment unresolved	Verify published table before issue
Section 1025.2.6.1	OCR/source ambiguity	Verify published clause
Section 1025.4	Corrupted luminance units in OCR	Raw text retained; no silent correction
Section 1027.6	Exception item 8 crosses pages	Verify published clause
Table 1030.13.2.1	8 concatenated row cells	Seat counts pending verification
Section 1030.17.3	OCR/source ambiguity	Verify published clause
Section 1030.17.4	OCR/source ambiguity	Verify published clause